

vasic-digital tmux — AGENTS.md

Base agent rules live at constitution/AGENTS.md and the constitution/Constitution.md it references. **READ THOSE FIRST.** This file extends them with project-specific rules; it never weakens them. Canonical project authority: Constitution.md — Project Articles §101–§109.

Fresh-conversation workflow: read CONTINUATION.md first (§0/§3/§8 mandatory), then Issues.md for OPEN/PARTIAL/BLOCKED/RUNNING.

MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

Operative rule. The bar for shipping is **not** "tests pass" but **"end users can actually use the feature."** Every PASS — test OR HelixQA Challenge — **MUST** carry positive captured runtime evidence that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-without-runtime PASS are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound **EQUALLY**.

This covenant is restated verbatim in every governance file at the consumer layer (CLAUDE.md, AGENTS.md, QWEN.md) so any tool that does not expand @imports still reads it. The same verbatim block lives upstream at constitution/Constitution.md §11.4, constitution/CLAUDE.md, and constitution/AGENTS.md. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: constitution/Constitution.md §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: Constitution.md §101.

Non-compliance is a release blocker regardless of context.

Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the tmx wrapper. Native dual-OS: each tmx new -s NAME spawns its own tmux server on the host with OS-native isolation — cgroup-v2 scope on Linux,

POSIX rlimit wrapper on macOS. Session shell is the operator's host shell with full PATH. No VM. macOS memory-cap gap documented honestly (XNU doesn't enforce RLIMIT_AS) — see docs/guide/README.md §5.6.

Critical base rules restated (for agents that don't follow @imports)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (file *content*, not existence). Universal §11.4 / Project §101.
- **FAIL-bluffs forbidden.** A test that exits FAIL for a script bug is fixed at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing language.** likely, probably, maybe, seems, appears are forbidden when reporting causes — prove with forensic evidence or mark UNCONFIRMED: / PENDING_FORENSICS:. Universal §11.4.6.
- **Operator-path coverage.** Every gate test exercises the SAME entry point an end-user invokes (tmx new -s X). Project §102.
- **Four-layer coverage.** Every defect lands pre-build gate + runtime test + HelixQA Challenge + paired mutation. Project §103.
- **Use the commit wrapper.** bash commit_all.sh "<msg>" only — no direct git add / git commit / git push on the main repo.
- **Never force-push.** Explicit per-operation authorization required.
- **CONTINUATION.md** kept in sync in every non-trivial commit. Universal §12.10.
- **60 % RAM cap** enforced per-session via TMX_MEM. Project §106.

Commands (exact)

Step	Command
Install build deps	bash scripts/install_deps.sh (Linux: root; macOS: brew)
Full pipeline	bash scripts/setup.sh
Build only / Verify only	bash scripts/setup.sh --build-only / --verify-only
Verification gate	bash scripts/verify.sh
Run all tests	bash scripts/tests/run_all.sh
Single test	bash scripts/tests/NN_*.sh (01..17)
Inheritance gate	bash scripts/tests/ test_constitution_inheritance.sh
Meta-test (paired mutation)	bash scripts/tests/ meta_test_false_positive_proof.sh
End-to-end automation	bash scripts/test_e2e.sh
Commit + push (github+gitlab)	bash commit_all.sh "message"
Per-session wrapper	tmx {new attach ls kill}

Structure

Path	Role
constitution/	HelixConstitution submodule — universal governance — do not modify
tmux/	upstream submodule (tag 3.6a) — do not modify
Containers/	vasic-digital cgroup helpers submodule
scripts/	build, verify, install, 17 tests + inheritance gate, challenges, templates
scripts/tmx	generated dispatcher (.gitignore'd — edit tmx.template)
scripts/tmux.conf.template	tmux config template (scrollback / copy-mode / clipboard)
commit_all.sh	only allowed push mechanism
Constitution.md	project authority (extends constitution/)
Issues.md / Fixed.md / CONTINUATION.md	open work / closed archive / live handoff

Project-specific agent rules

Allowed CLI tools

- git, gh, glab (releases), bash, the project's own scripts.
- macOS: Homebrew (brew) for build deps. Linux: distro package manager via install_deps.sh.

Forbidden in this project

- Editing tmux/, constitution/, or generated files (scripts/tmx, tmux/build*/).
- Direct git push on the main repo — use commit_all.sh.
- CI/CD pipeline files (.github/workflows/, .gitlab-ci.yml).
- HTTPS Git remotes — SSH only.
- Advancing the tmux/ submodule off tag 3.6a without an explicit documented decision (Project §107).

Project-specific workflow

- **Test-interrupt-on-discovery** (§103): a defect found mid-cycle STOPS the cycle — fix root cause, land four layers, rebuild, repeat.
- **Topology dispatch** (§104): tests detect host topology and SKIP-with-reason; never silently degrade.
- **Destructive tests** (12/13/14): require TMX_TEST_DESTRUCTIVE=1.
- **Releases**: bump VERSION (version= + versionCode= together) + CHANGELOG.md entry; tag + gh release create + glab release create.

Project overrides of universal rules

None.

§11.4.87–98 — newly-inherited anchors (constitution 6828ff2)

Twelve new universal anchors landed in constitution/Constitution.md at commit 6828ff2 (2026-05-28). They apply unconditionally per §11.4.35. Short-form mirrors follow.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch

Inherited from constitution/Constitution.md §11.4.87. Continue work until all of: Issues.md has zero non-terminal items, CONTINUATION.md §3 active-work empty, no in-flight subagent, no external dependency in-flight. Every closure carries §11.4.4(b) four-layer coverage + captured-evidence per §11.4.5/§11.4.69. **Canonical authority:** constitution/Constitution.md §11.4.87.

§11.4.88 — Background-push: commit-flock release immediate, push detached

Inherited from constitution/Constitution.md §11.4.88. commit_all.sh releases its commit flock as soon as git commit returns 0; push runs detached via nohup ... & + disown. Per-remote flock serialises same-remote concurrent pushes. Only escape: --sync-push for §11.4.41 force-push paths. **Canonical authority:** constitution/Constitution.md §11.4.88.

§11.4.89 — Background test execution

Inherited from constitution/Constitution.md §11.4.89. Any test

```
30 s (run_all.sh, meta_test_false_positive_proof.sh,
test_e2e.sh) MUST run detached via nohup ... > <log> 2>&1 & +
disown; main stream returns to the §11.4.42 priority queue. Canonical
authority: constitution/Constitution.md §11.4.89.
```

§11.4.90 — Obsolete status + per-item obsolescence audit

Inherited from constitution/Constitution.md §11.4.90. New §11.4.15 Status terminal value Obsolete (→ Fixed.md) with mandatory **Obsolete–Details:** (Since + Reason from closed vocabulary + Superseding-item + Triple-check evidence). Colorizer marks rows light-gray + strikethrough. **Canonical authority:** constitution/Constitution.md §11.4.90.

§11.4.91 — Summary-doc clarity

Inherited from constitution/Constitution.md §11.4.91. Every summary one-liner ≥ 6 words OR ≥ 40 chars, naming SUBJECT + PROBLEM/GOAL. Anti-patterns forbidden: section labels, bare metadata fragments, §-letter alone. Generators extract from H1/H2 heading per §11.4.54 ATM-NNN convention. **Canonical authority:** constitution/Constitution.md §11.4.91.

§11.4.92 — Multi-pass change-evaluation discipline

Inherited from constitution/Constitution.md §11.4.92. 5 passes before commit: (1) main task evidence; (2) regression blast radius; (3) cross- feature interaction; (4) deep research + CodeGraph; (5) anti-bluff confirmation. Trivial exemption only for zero-source-touch commits that cite it. **Canonical authority:** constitution/Constitution.md §11.4.92.

§11.4.93 — SQLite-backed single source of truth for workable items

Inherited from constitution/Constitution.md §11.4.93. Text trackers become views of docs/workable_items.db. Go binary cmd/workable-items/ provides md-to-db / db-to-md / diff / validate. Round-trip is byte-identical within whitespace tolerance. commit_all.sh refuses non-empty diff. **Canonical authority:** constitution/Constitution.md §11.4.93.

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode

Inherited from constitution/Constitution.md §11.4.94. Idle only when genuinely externally blocked OR operator STOP OR §12 host-safety. Before any wake/sleep, survey parallel-work feasibility per §11.4.42 / §11.4.72 / §11.4.87 and dispatch non-contending items via §11.4.20 / §11.4.70. **Canonical authority:** constitution/Constitution.md §11.4.94.

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored

Inherited from constitution/Constitution.md §11.4.95. Amends §11.4.93: docs/workable_items.db is authoritative source data, TRACKED in git, never gitignored. Every mutation stages + commits + pushes alongside the MD regen. WAL-checkpoint before commit-stage. **Canonical authority:** constitution/Constitution.md §11.4.95.

§11.4.96 — Safe-parallel-work-with-long-build catalogue

Inherited from constitution/Constitution.md §11.4.96. Operational catalogue classifies parallel work as SAFE (docs/scripts/tests/submodule pushes, research, DB ops, subagent dispatch) vs UNSAFE (destructive git ops, mass deletes, host-safety breaches). Conductor consults catalogue at every pause point. **Canonical authority:** constitution/Constitution.md §11.4.96.

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence

Inherited from constitution/Constitution.md §11.4.97. Idle minute during which work could progress and is not genuinely blocked = §11.4.97 violation. Emit 1-line operator-facing update on every commit landed / subagent return / anchor landing / evidence acquisition / milestone closure. **Canonical authority:** constitution/Constitution.md §11.4.97.

§11.4.98 — Full-Automation Anti-Bluff: live tests re-runnable without manual intervention

Inherited from constitution/Constitution.md §11.4.98. Every test MUST be fully self-driving end-to-end and emit PASS/FAIL/SKIP-with-reason without further human action after startup. Only exception: one-time credential bootstrap OUTSIDE test execution. Re-runability proof = PASS at -count=3 consecutive automated invocations. **Canonical authority:** constitution/Constitution.md §11.4.98.

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate

Inherited from constitution/Constitution.md §11.4.99 (commit 9e3bcc5 ff 2026-05-28). Before committing operator-facing instruction docs (guides, manuals, setup walkthroughs, troubleshooting cookbooks, API how-tos), fetch LATEST official online docs via WebFetch / MCP / direct browsing (NOT training data). Cross-reference every instruction. ## Sources verified <date> section in doc + Sources verified <date>: <urls> footer in commit message MANDATORY. Re-verification cadence: ≤6 mo general; ≤90 d for risk-classified (messengers, cloud, payments, LLM, code-host, package mgrs). Stale → §11.4.90 Obsolete after 30-day grace. **Canonical authority:** constitution/Constitution.md §11.4.99.